FLIGHT PERFORMANCE ANALYSIS OF A UAV

Prepared by
Talha Şen

Field: Aerospace Engineering
Focus: Aerodynamics and Flight Performance

Tools Used:
XFLR5, Microsoft Excel

Project Type:
Independent Engineering Study

Date: 2026

ABSTRACT

This study presents a comprehensive flight performance analysis of a small-scale fixed-wing unmanned aerial vehicle (UAV). The objective of the study is to evaluate the aerodynamic behavior and overall performance characteristics of the UAV under various flight conditions.

The analysis includes the investigation of aerodynamic properties using the NACA 2412 airfoil, where lift, drag, and aerodynamic efficiency characteristics were obtained through XFLR5 simulations. Based on these results, the stall behavior of the aircraft was analyzed, and the stall speed was determined.

Furthermore, drag analysis was conducted by separating the total drag into parasitic and induced components. Using this model, the drag force was calculated for a range of flight velocities. The power required for level flight was then obtained as a function of drag and velocity, and it was compared with the available power to determine the operational flight limits.

In addition, throttle analysis was performed to evaluate engine loading at different speeds, while climb performance was assessed using excess power and rate of climb (ROC) calculations. Endurance and range analyses were also carried out to determine the optimal flight speeds for maximum flight duration and maximum distance, respectively.

Finally, the flight envelope and maneuver envelope (V–n diagram) of the UAV were constructed. Key performance speeds such as stall speed, safe speed, maneuver speed, maximum endurance speed, maximum range speed, and maximum flight speed were identified.

The results show that the UAV achieves maximum endurance at approximately 15 m/s and maximum range at around 22 m/s, while the safe operational velocity range lies between approximately 13 m/s and 33 m/s. Overall, this study provides a detailed understanding of the aerodynamic and performance limits of the UAV and identifies the optimal operating conditions for efficient and safe flight.

İçindekiler

1. Introduction 1

2. UAV Geometry and Basic Parameters 2

2.1 Aspect Ratio Calculation 2

2.2 Wing Loading Calculation 3

2.3 Influence of Geometric Parameters on Flight Performance 3

3. Aerodynamic Characteristics (Airfoil Analysis) 4

4. Stall Analysis 7

5. Drag Analysis 9

5.1 Selection of Drag Model 10

5.2 Preparation of Geometric Data 10

5.3 Obtaining Aerodynamic Data from Airfoil Analysis 10

5.4 Estimation of $CD0$ 11

5.5 Calculation of the k Coefficient for Induced Drag 12

5.6 Determination of CL at the Selected Flight Condition 13

5.7 Calculation of the Induced Drag Coefficient 13

5.8 Determination of the Total Drag Coefficient 14

5.9 Calculation of the Actual Drag Force 14

5.10 Repetition of Calculations for Different Velocities 15

5.11 Drag Interpretation Based on L/D Ratio 17

6. Power Required Analysis 17

6.1 Power Required vs Power Available 19

6.2 Throttle Analysis 20

6.3 Climb Performance Analysis 22

7. Endurance Analysis 24

8. Range Analysis 26

9. Flight Envelope 28

10. Conclusion 30

1. Introduction

Unmanned Aerial Vehicles (UAVs) have become increasingly important in both civilian and military applications due to their versatility, operational efficiency, and their ability to perform missions without putting human life at risk. Today, UAVs are widely used in a variety of missions such as surveillance, reconnaissance, environmental monitoring, mapping, and communication. In addition, UAVs also play a significant role in military operations, where some UAV systems are developed as armed platforms capable of target detection and precision strike missions. In order to successfully carry out these missions, it is essential to have a thorough understanding of the aerodynamic characteristics and flight performance of the aircraft.

Flight performance analysis is one of the fundamental components of aircraft design and operation. This analysis makes it possible to examine the behavior of an aircraft under different flight conditions and helps determine important performance parameters such as stall speed, drag characteristics, power required, endurance, range, and the safe flight limits of the aircraft. These parameters play a critical role in evaluating the efficiency and operational capabilities of a UAV.

The aerodynamic performance of an aircraft largely depends on its geometric configuration. Parameters such as wing area, wingspan, aspect ratio, and wing loading determine how efficiently the aircraft can generate lift and how much drag is produced during flight. In particular, induced drag and parasitic drag are important factors that significantly affect the overall performance of the UAV and the power required during flight.

The objective of this study is to perform a flight performance analysis of a small-scale fixed-wing UAV. Within the scope of this study, the geometric properties of the aircraft, aerodynamic characteristics, stall behavior, drag components, power required for steady flight, endurance performance, range capability, and the safe flight limits of the aircraft will be examined.

By analyzing these parameters, the study aims to provide a comprehensive evaluation of the UAV's flight performance and aerodynamic efficiency. The results obtained from this analysis will help identify the most efficient operating conditions of the UAV and provide important insights into the overall performance characteristics of the aircraft.

2. UAV Geometry and Basic Parameters

In this section, the fundamental geometric and physical parameters of the unmanned aerial vehicle (UAV) used in the flight performance analysis are defined. Accurate determination of these parameters is essential for calculating lift, drag, and overall flight performance characteristics. The geometric configuration of the wing and the mass properties of the aircraft play a critical role in determining the aerodynamic behavior of the UAV.

For the purpose of this study, a small fixed-wing UAV configuration is considered as the reference aircraft.

Table 1. UAV Basic Parameters

Parameter	**Symbol**	**Value**
Mass	m	8 kg
Weight	W	78.48 N
Wing Area	S	0.8 m²
Wingspan	b	2.4 m
Air Density	ρ	1.225 kg/m³
Airfoil	—	NACA 2412

The weight of the UAV is calculated using the mass and gravitational acceleration:

$$W = m.g$$

$$W = 8 \times 9.81 = 78.48\ N$$

This value will be used in the aerodynamic and flight performance calculations in the following sections.

2.1 Aspect Ratio Calculation

One of the most important parameters affecting aerodynamic efficiency is the **aspect ratio (AR)** of the wing. The aspect ratio is defined as the square of the wingspan divided by the wing area.

$$AR = \frac{b^2}{S}$$

For the selected UAV configuration:

$$AR = \frac{2.4^2}{0.8}$$

$$AR = \frac{5.76}{0.8}$$

$$AR = 7.2$$

This value indicates that the UAV has a moderately high aspect ratio wing configuration. Wings with higher aspect ratios generally produce lower induced drag and therefore improve aerodynamic efficiency.

2.2 Wing Loading Calculation

Wing loading is defined as the ratio of the aircraft weight to the wing area and is an important parameter affecting flight performance.

$$Wing\ Loading = \frac{W}{S}$$

$$Wing\ Loading = \frac{78.48}{0.8}$$

$$Wing\ Loading = 98.1\ N/m^2$$

This value can be approximately expressed as:

$$10\ kg/m^2$$

Lower wing loading values are typically associated with lower stall speeds, improved low-speed flight performance, and increased endurance. For this reason, many surveillance and reconnaissance UAVs are designed with relatively low wing loading.

2.3 Influence of Geometric Parameters on Flight Performance

The geometric parameters defined above provide important insights into the expected flight behavior of the UAV.

- The **aspect ratio** significantly affects the induced drag characteristics of the wing. Higher aspect ratios generally reduce induced drag and improve aerodynamic efficiency.
- The **wing loading** directly influences stall speed and minimum safe flight speed.
- The **wing area** determines the lifting capability of the aircraft.

These parameters form the basis for the subsequent analyses in this study, including stall performance, drag characteristics, power required, endurance, and range calculations.

3. Aerodynamic Characteristics (Airfoil Analysis)

In this section, the aerodynamic behavior of the selected airfoil is analyzed. The aerodynamic characteristics of an airfoil directly influence the lift generation capability, drag characteristics, and overall aerodynamic efficiency of an aircraft. Therefore, analyzing the airfoil performance is an essential part of the flight performance evaluation.

In this study, the **NACA 2412 airfoil** was selected for the aerodynamic analysis. The NACA 2412 airfoil is widely used in small aircraft and unmanned aerial vehicles due to its favorable lift characteristics and relatively low drag.

The aerodynamic analysis of the airfoil was performed using the **XFLR5** software. XFLR5 is a widely used aerodynamic analysis tool for evaluating airfoils and wing configurations operating at low Reynolds numbers. In this analysis, key aerodynamic parameters such as lift coefficient, drag coefficient, and aerodynamic efficiency were investigated.

The analysis was performed under the following conditions:

Reynolds number:

$$Re = 300000$$

Mach number:

$$M = 0$$

Critical amplification factor:

$$N_{crit} = 9$$

Angle of attack range:

$$-4^\circ \leq \alpha \leq 18^\circ$$

To evaluate the aerodynamic behavior of the airfoil, the following plots were obtained:

- **CL vs α curve (Lift Curve)**
- **CD vs α curve**
- **CL/CD vs α curve (Aerodynamic efficiency)**
- **CL vs CD curve (Drag Polar)**

The **CL vs α** curve shows that the lift coefficient increases with increasing angle of attack.

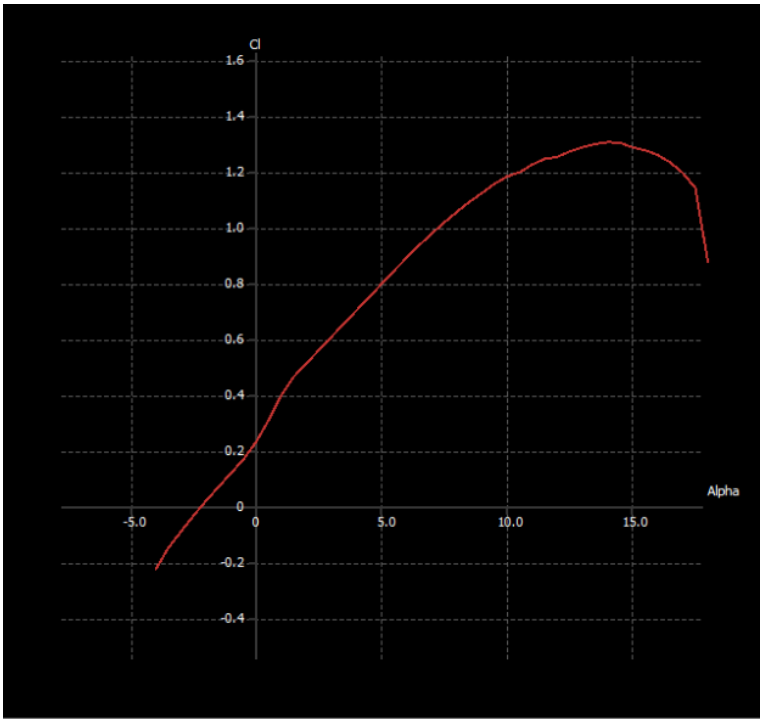

Figure 1. *Variation of lift coefficient (CL) with respect to the angle of attack (α) for the NACA 2412 airfoil.*

The lift coefficient increases approximately linearly with increasing angle of attack and reaches its maximum value at around **15°**. Beyond this point, a decrease in the lift coefficient is observed, indicating the onset of stall.

As a result of the analysis, the maximum lift coefficient is determined as:

$$C_{L_{max}} \approx 1.3$$

This value is obtained at an angle of attack of approximately:

$$\alpha \approx 15^{\circ}$$

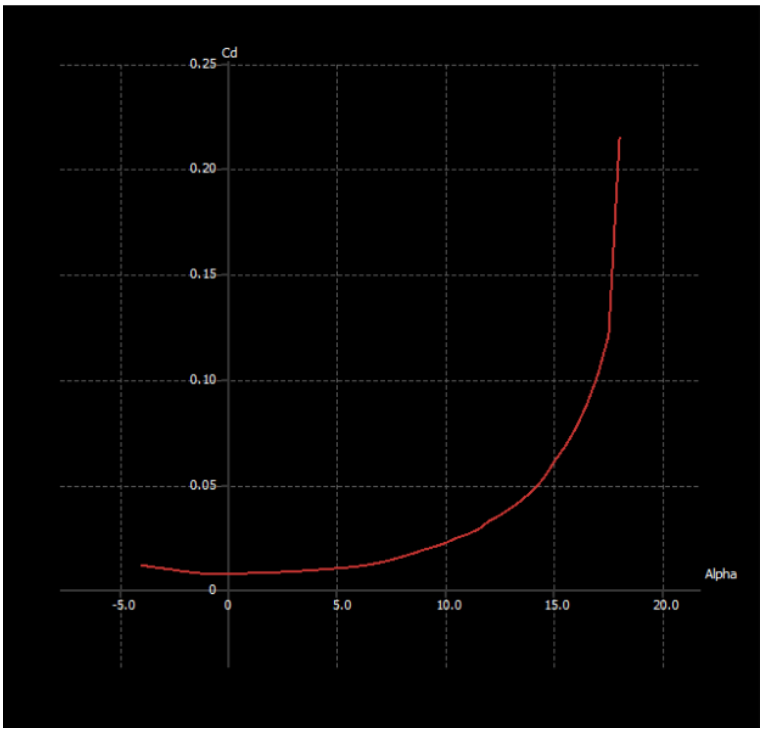

Figure 2. *Variation of drag coefficient (CD) with respect to the angle of attack (α).*

The drag coefficient increases gradually with increasing angle of attack. At higher angles of attack, the drag rises more rapidly due to flow separation and increased aerodynamic resistance. The aerodynamic efficiency was evaluated using the **CL/CD vs α** curve.

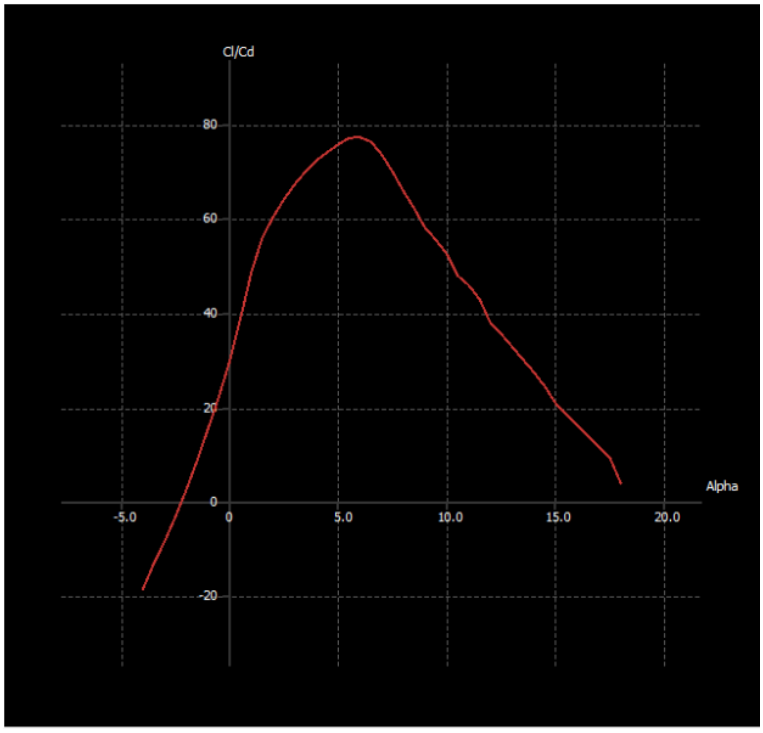

Figure 3*. Variation of lift-to-drag ratio (CL/CD) with respect to the angle of attack (α).*

The maximum lift-to-drag ratio represents the most aerodynamically efficient operating condition of the airfoil. As a result of the analysis, this value is determined as:

$$(CL/CD)_{max} \approx 78$$

This value occurs at an angle of attack of approximately:

$$\alpha \approx 6^{\circ}$$

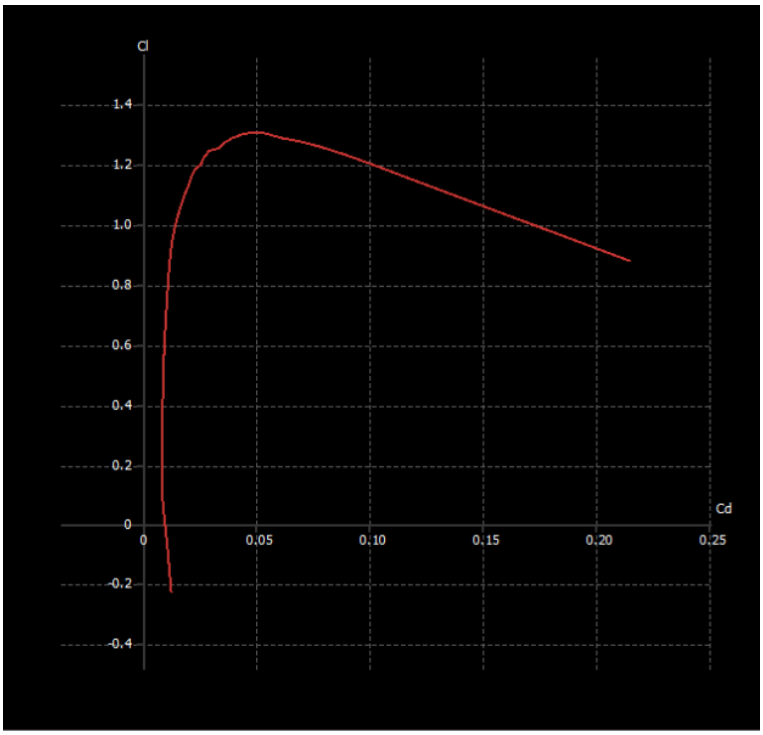

Figure 4. *Drag polar (CL–CD relationship) of the NACA 2412 airfoil*

The drag polar curve shows the relationship between the lift coefficient and the drag coefficient. This curve plays an important role in understanding the aerodynamic performance of the airfoil and is widely used in flight performance calculations.

The obtained results indicate that the selected airfoil provides suitable aerodynamic characteristics for small-scale unmanned aerial vehicles. Furthermore, the parameters obtained from this analysis will be used as the basis for subsequent calculations, including stall analysis, drag analysis, and flight performance evaluation.

4. Stall Analysis

Stall is one of the most critical aerodynamic phenomena affecting the safe operation of an aircraft. It occurs when the angle of attack exceeds a certain critical value, leading to flow separation over the airfoil and a significant loss of lift.

Based on the aerodynamic analysis of the NACA 2412 airfoil, the maximum lift coefficient was determined as:

$$C_{L_{max}} \approx 1.3$$

The corresponding stall angle was found to be approximately:

$$\alpha_{stall} \approx 15^\circ$$

The stall speed of the UAV can be calculated using the following equation:

$$V_{stall} = \sqrt{\frac{2W}{\rho S C_{L_{max}}}}$$

where:

- W is the weight of the aircraft
- ρ is the air density
- S is the wing area
- $C_{L_{max}}$ is the maximum lift coefficient

For the given UAV:

$$W = 78.48\ N$$

$$\rho = 1.225\ kg/m^3$$

$$S = 0.8\ m^2$$

$$C_{L_{max}} = 1.3$$

Substituting these values:

$$V_{stall} = \sqrt{\frac{2 \times 78.48}{1.225 \times 0.8 \times 1.3}}$$

$$V_{stall} = \sqrt{123.2}$$

$$V_{stall} \approx 11.1\ m/s$$

This result indicates that the UAV is expected to stall at approximately **11.1 m/s**. In practical flight conditions, the aircraft should operate above this speed with a safety margin.

In real flight conditions, aircraft do not operate at stall speed. A safety factor is applied to ensure safe flight, and the minimum safe flight speed is generally taken as:

$$V_{safe} \approx 1.2 \times V_{stall}$$

For this UAV:

$$V_{safe} \approx 13.3\, m/s$$

The relatively low stall speed obtained in this study is directly related to the low wing loading calculated in Section 2. Lower wing loading allows the aircraft to generate sufficient lift at lower flight speeds. This improves low-speed performance, reduces the risk of stall, and enables the aircraft to maintain better control during flight.

In addition, low wing loading positively affects takeoff and landing performance by allowing operations over shorter distances. This is a significant advantage, especially for small-scale unmanned aerial vehicles.

The low stall speed enables the UAV to perform stable and safe flight at low speeds in missions such as surveillance, reconnaissance, and environmental monitoring. In such operations, the ability to remain over a specific area for extended periods and maintain control at low speeds is critical.

In conclusion, the low wing loading and the resulting low stall speed provide an important design advantage by enhancing the operational efficiency and directly improving the mission performance of the UAV.

5. Drag Analysis

Drag force is the aerodynamic force acting opposite to the direction of motion during flight and is one of the most important parameters affecting aircraft performance. The power required, aerodynamic efficiency, endurance, and range capability of an aircraft are strongly influenced by the drag force.

Total drag generally consists of two main components: parasitic drag and induced drag. Parasitic drag results from the interaction of the aircraft surfaces with the airflow, while induced drag arises as a consequence of lift generation. The variation of these components under different flight conditions plays a critical role in determining the efficient operating region of the aircraft.

In this section, the drag characteristics of the aircraft will be analyzed using an analytical model and aerodynamic data. The components of the drag coefficient will be evaluated separately, the drag force will be calculated for a representative flight condition, and the results will be interpreted in terms of aerodynamic efficiency.

5.1 Selection of Drag Model

Accurate modeling of drag force is essential in flight performance analysis. In this study, the total drag coefficient is expressed using the following analytical model:

$$C_D = C_{D0} + kC_L^2$$

In this model, the total drag coefficient is defined as the sum of two main components. The term C_{D0} represents the parasitic drag coefficient at zero lift condition. This component accounts for the drag generated by the interaction of the fuselage, wing surfaces, and other external elements with the airflow, and it generally increases with flight speed.

The second term, kC_L^2 , represents the induced drag component. This type of drag is directly related to lift generation and is caused by wingtip vortices. It becomes more significant at low speeds and high lift conditions.

This model clearly describes how the drag coefficient varies with the lift coefficient and is widely used in flight performance analysis. Therefore, it is adopted as the fundamental drag model in this study.

5.2 Preparation of Geometric Data

In order to perform the drag analysis, the fundamental geometric and physical parameters of the aircraft must first be defined. These parameters are directly used in the calculation of the drag coefficient and the resulting drag force.

The basic parameters used in this study are given below:

- Weight: $W = 78.48\ N$
- Wing area: $S = 0.8\ m^2$
- Wingspan: $b = 2.4\ m$
- Aspect ratio: $AR = 7.2$
- Air density: $\rho = 1.225\ kg/m^3$

These parameters are based on the UAV geometry defined in the previous section and will be used in the subsequent stages of the drag analysis. In particular, wing area, aspect ratio, and air density are critical for the calculation of total drag, induced drag, and drag force.

5.3 Obtaining Aerodynamic Data from Airfoil Analysis

In the next stage of the drag analysis, it is necessary to determine the aerodynamic data of the selected airfoil. For this purpose, the results previously obtained from the airfoil analysis performed using **XFLR5** were used. The airfoil analysis provides the main aerodynamic parameters such as lift coefficient, drag coefficient, lift-to-drag ratio, and drag polar characteristics.

In this study, the following aerodynamic plots of the **NACA 2412** airfoil were examined:

- $C_L - \alpha \ curve$
- $C_D - \alpha \ curve$
- $C_L/C_D - \alpha \ curve$
- $C_L - C_D \ curve(drag \ polar)$

The data obtained from these plots made it possible to determine the main parameters used in the drag analysis. In particular, the drag polar curve provided an important reference for estimating C_{DO} and interpreting the overall drag behavior by showing the relationship between lift coefficient and drag coefficient. In addition, the maximum C_L/C_D value was used to identify the most aerodynamically efficient operating condition of the aircraft.

Therefore, the aerodynamic data obtained from the airfoil analysis constitute the main input for the numerical and physical interpretation of the drag analysis.

5.4 Estimation of C_{D0}

One of the important parameters in determining the total drag coefficient is C_{D0} , which represents the drag coefficient at the zero-lift condition. This parameter corresponds to the baseline aerodynamic drag of the aircraft in the absence of lift generation.

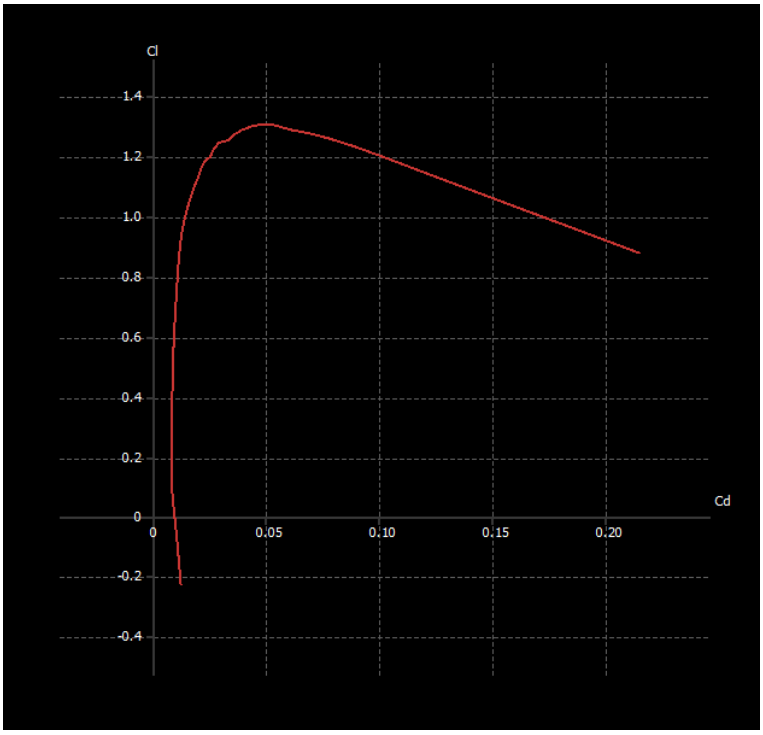

Figure 5. *Drag polar (CL–CD relationship) of the NACA 2412 airfoil.*

The value of C_{D0} can be approximately estimated using the drag polar curve(*Figure 5)*. By examining the drag coefficient near the condition where $C_L = 0$, it can be observed that the curve(*Figure 5)* intersects the C_D axis at approximately:

$$C_{D0} \approx 0.007$$

This value represents the level of parasitic drag caused by the interaction of the fuselage, wing surface, and other external geometric components with the airflow. Therefore, C_{D0} will be used as one of the fundamental components in the subsequent total drag calculations.

5.5 Calculation of the *k* Coefficient for Induced Drag

The second term in the total drag model represents the induced drag component. The magnitude of this component depends on the coefficient ￼ in the following expression:

$$C_{D_i} = kC_L^2$$

The coefficient ￼ is determined by the geometric characteristics and aerodynamic efficiency of the wing. It can be calculated using:

$$k = \frac{1}{\pi ARe}$$

where:

- AR : aspect ratio
- e : Oswald efficiency factor

In this study, the aspect ratio previously calculated for the UAV was taken as:

$$AR = 7.2$$

The Oswald efficiency factor was assumed as:

$$e \approx 0.8$$

which is a commonly used approximate value for small fixed-wing UAVs.

Substituting these values:

$$k = \frac{1}{\pi \times 7.2 \times 0.8}$$

$$k \approx 0.055$$

was obtained.

This result represents the fundamental coefficient used in the calculation of the induced drag component. Therefore, in the following steps, the induced drag coefficient for a given lift coefficient will be determined using this value.

5.6 Determination of C_L at the Selected Flight Condition

In order to perform the drag analysis, the lift coefficient (C_L) corresponding to a specific flight condition must be determined. In steady level flight, the lift force is equal to the weight of the aircraft:

$$L = W$$

The lift force is expressed as:

$$L = \frac{1}{2}\rho V^2 S C_L$$

Combining these equations, the lift coefficient can be written as:

$$C_L = \frac{2W}{\rho V^2 S}$$

In this study, the flight speed for a representative cruise condition was taken as:

$$V = 15\, m/s$$

The following values were used:

$$W = 78.48\, N\,, \quad \rho = 1.225\, kg/m^3, \quad S = 0.8\, m^2$$

Substituting these values:

$$C_L = \frac{2 \times 78.48}{1.225 \times 15^2 \times 0.8}$$

$$C_L \approx 0.71$$

This value represents the lift coefficient required to balance the weight of the UAV at the selected cruise speed and will be used in subsequent drag calculations.

5.7 Calculation of the Induced Drag Coefficient

In the previous step, the lift coefficient required for the selected flight condition was determined as:

$$C_L \approx 0.71$$

In addition, the value of k to be used in the induced drag calculation was obtained in the previous section as:

$$k \approx 0.055$$

The induced drag coefficient is calculated using the following expression:

$$C_{D_i} = kC_L^2$$

Substituting the values:

$$C_{D_i} = 0.055 \times (0.71)^2$$

$$C_{D_i} = 0.055 \times 0.5041$$

$$C_{D_i} \approx 0.0277$$

This result represents the induced drag coefficient generated due to lift production at the selected cruise condition. Since induced drag becomes more significant at low speeds and higher lift requirements, it should be considered as an important component in flight performance analysis.

5.8 Determination of the Total Drag Coefficient

The total drag coefficient is expressed as the sum of the parasitic drag and induced drag components:

$$C_D = C_{D0} + C_{D_i}$$

In the previous steps, the zero-lift drag coefficient was determined as:

$$C_{D0} \approx 0.007$$

and the induced drag coefficient for the selected flight condition was calculated as:

$$C_{D_i} \approx 0.0277$$

Substituting these values:

$$C_D = 0.007 + 0.0277$$

$$C_D \approx 0.0347$$

This result represents the total drag coefficient of the UAV at the selected cruise condition. Since it includes both parasitic and induced drag effects, it serves as a fundamental parameter for subsequent flight performance calculations.

5.9 Calculation of the Actual Drag Force

In the previous step, the total drag coefficient for the selected cruise condition was determined as:

$$C_D \approx 0.0347$$

The actual drag force is calculated using the following equation:

$$D = \frac{1}{2}\rho V^2 S C_D$$

where:

- $\rho = 1.225\ kg/m^3$ is the air density,
- $V = 15\ m/s$ is the cruise speed,
- $S = 0.8\ m^2$ is the wing area,
- $C_D = 0.0347$ İs the total drag coefficient.

Substituting the values:

$$D = \frac{1}{2} \times 1.225 \times 15^2 \times 0.8 \times 0.0347$$

$$D = 0.5 \times 1.225 \times 225 \times 0.8 \times 0.0347$$

$$D \approx 3.83\ N$$

This result indicates that the UAV is subjected to an actual drag force of approximately **3.83 N** under the selected cruise condition. This value represents the aerodynamic resistance that must be overcome during flight and forms the basis for the power required analysis in the next section.

5.10 Repetition of Calculations for Different Velocities

To evaluate the aerodynamic behavior of the UAV under different flight conditions, drag calculations were repeated for a range of velocities between **10–35 m/s**. The lift coefficient for each speed was calculated using:

$$C_L = \frac{2W}{\rho V^2 S}$$

The induced drag coefficient and total drag coefficient were then determined using:

$$C_{D_i} = kC_L^2$$

$$C_D = C_{D0} + C_{D_i}$$

Finally, the drag force was calculated as:

$$D = \frac{1}{2}\rho V^2 S C_D$$

The results are presented in **Table 2**, and the variation of drag force with velocity is shown in **Figure 6**.

Table 2. *Calculated values of C_L, C_{D_i}, C_D and drag force for different flight speeds*

V	CL	CDi	CD	D
10	1,601632653	0,141087494	0,148087494	7,256287184
12	1,112244898	0,068039879	0,075039879	5,294813878
15	0,711836735	0,027869135	0,034869135	3,844322082
18	0,494331066	0,013439976	0,020439976	3,245050612
20	0,400408163	0,008817968	0,015817968	3,100321796
22	0,330915837	0,006022791	0,013022791	3,088485121
25	0,256261224	0,00361184	0,01061184	3,249875949
28	0,204289879	0,00229539	0,00929539	3,570916835
30	0,177959184	0,001741821	0,008741821	3,85514302
35	0,130745523	0,000940192	0,007940192	4,766099974

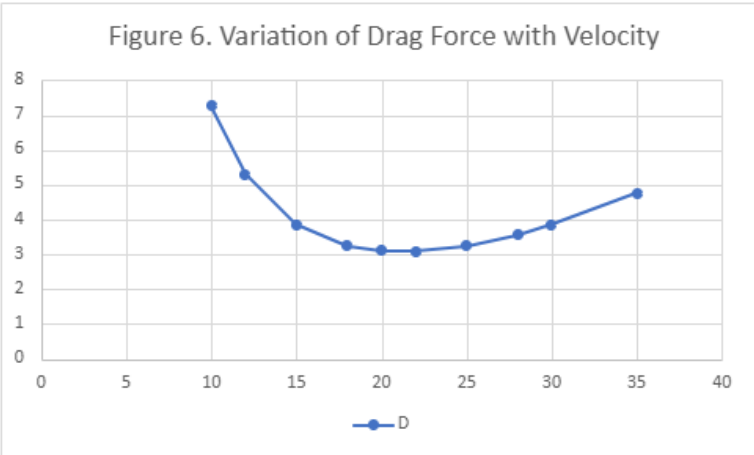

Figure 6. *Variation of drag force with velocity*

As shown in Table 2 and Figure 6, the drag force initially decreases with increasing velocity and then increases after a certain point. This behavior is due to the variation of induced and parasitic drag components with speed.

At low velocities, higher lift is required to balance the aircraft weight, which results in higher induced drag. As velocity increases, the required lift coefficient decreases, leading to a reduction in induced drag.

However, at higher velocities, parasitic drag becomes dominant and causes the total drag to increase again. As a result, the drag force exhibits a U-shaped variation with velocity.

According to the results, the minimum drag occurs approximately at:

$$V \approx 20 - 22\, m/s$$

This speed represents the condition at which the aircraft experiences the lowest aerodynamic drag. This speed is considered a critical parameter for range optimization and efficient flight planning.

5.11 Drag Interpretation Based on L/D Ratio

As a result of the aerodynamic analysis, the maximum lift-to-drag ratio was found to be:

$$(C_L/C_D)_{max} \approx 78$$

This value indicates that the aircraft has high aerodynamic efficiency.

The maximum L/D ratio represents the flight condition at which the aircraft can generate the required lift with the minimum drag. Therefore, this point is considered the most aerodynamically efficient operating condition.

In this study, the maximum L/D value was observed at approximately:

$$\alpha \approx 6^\circ$$

This angle represents the most efficient angle of attack for the aircraft.

Although the maximum L/D point and the minimum drag point are close to each other, they are not identical. The minimum drag point corresponds to the lowest total drag, whereas the maximum L/D point represents the most favorable balance between lift and drag.

Therefore, the maximum L/D condition is particularly important for range optimization and long-endurance missions. It provides a significant advantage for UAV operations such as surveillance and reconnaissance, where efficient flight performance is critical.

6. Power Required Analysis

In flight performance analysis, determining the power required for an aircraft to maintain flight at a given velocity is of great importance. This analysis reveals the energy demand of the aircraft at different flight conditions and helps identify optimal operating points.

Power required represents the power needed to overcome aerodynamic drag. Therefore, power analysis is directly dependent on the drag analysis and serves as its continuation.

The required power is calculated as the product of drag force and velocity:

$$P = D \times V$$

where:

- P : Power required (Watt)
- D : Drag force (Newton)
- V : Flight velocity (m/s)

In this study, the drag values previously calculated for different velocities were used to determine the power required at each speed. The results are presented in **Table 3**, and the variation of power with velocity is shown in **Figure 7**.

Table 3. *Calculated values of C_L, C_{D_i}, C_D drag force, and power required for different flight speeds*

V	CL	CDi	CD	D	P (W)
10	1,601632653	0,141087494	0,148087494	7,256287184	72,56287184
12	1,112244898	0,068039879	0,075039879	5,294813878	63,53776653
15	0,711836735	0,027869135	0,034869135	3,844322082	57,66483122
18	0,494331066	0,013439976	0,020439976	3,245050612	58,41091102
20	0,400408163	0,008817968	0,015817968	3,100321796	62,00643592
22	0,330915837	0,006022791	0,013022791	3,088485121	67,94667265
25	0,256261224	0,00361184	0,01061184	3,249875949	81,24689873
28	0,204289879	0,00229539	0,00929539	3,570916835	99,98567137
30	0,177959184	0,001741821	0,008741821	3,85514302	115,6542906
35	0,130745523	0,000940192	0,007940192	4,766099974	166,8134991

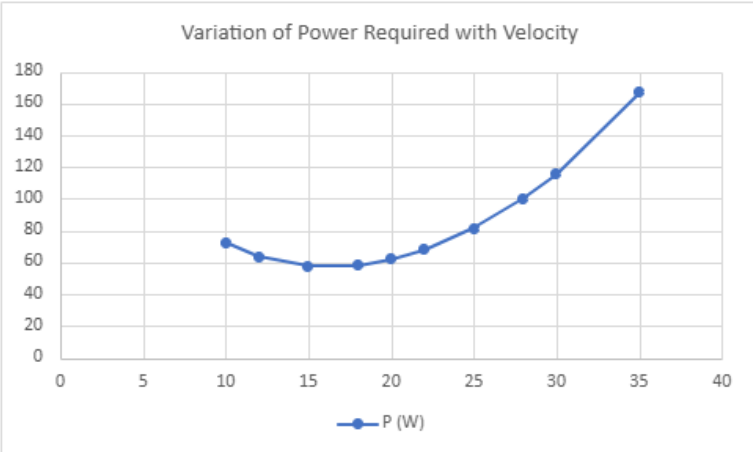

Figure 7. *Variation of Power Required with Velocity*

As shown in Table 3 and Figure 7, the required power initially decreases with increasing velocity and then increases after a certain point. This behavior is directly related to the variation of drag with velocity.

At low speeds, higher lift is required to balance the aircraft weight, resulting in increased induced drag and therefore higher power consumption. As velocity increases, induced drag decreases, leading to a reduction in required power.

However, at higher velocities, parasitic drag increases significantly, causing the required power to rise again.

According to the results, the minimum power occurs approximately at:

$$V \approx 15\, m/s$$

This speed represents the condition at which the aircraft achieves maximum endurance, meaning it can remain airborne for the longest duration.

6.1 Power Required vs Power Available

In flight performance analysis, it is essential to compare the power required for flight with the power available from the propulsion system. This comparison helps determine the flight limits and operational speed range of the aircraft.

In this study, the calculated power required values were compared with the assumed power available.

The power required was calculated using:

$$P_{req} = D \times V$$

The power available was assumed to be approximately constant for the UAV:

$$P_{avail} \approx 150\, W$$

The calculated power required values were then compared with this constant available power. The results are presented in **Figure 8**.

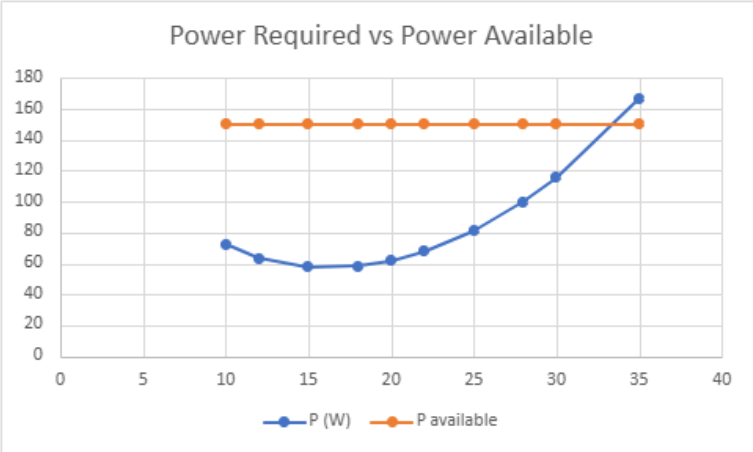

Figure 8. *Power Required vs Power Available*

As shown in Figure 8, the power required initially decreases with increasing velocity and then increases after a certain point. The power available is represented as a constant horizontal line.

At low and moderate speeds, the power required remains below the power available, indicating that the UAV can sustain steady level flight in this range.

However, as velocity increases, the power required approaches the available power at approximately **33–34 m/s**. Beyond this point:

$$P_{req} > P_{avail}$$

which means that steady level flight is not possible.

Therefore, the maximum level flight speed of the UAV is limited to a value slightly below the intersection point of the power required and power available curves.

6.2 Throttle Analysis

Throttle analysis is used to determine how much engine power is required for the aircraft to sustain flight at a given velocity. This analysis provides insight into engine loading and performance limits during flight.

The throttle value is defined as the ratio of required power to available power:

$$Throttle = \frac{P_{req}}{p_{avail}}$$

In this study, the available power was assumed to be approximately **150 W**.

Table 4. *Aerodynamic coefficients, drag, power required, power available, and throttle values for different flight speeds*

V	CL	CDi	CD	D	P (W)	P avail	Throttle
10	1,601632653	0,141087494	0,148087494	7,256287184	72,56287184	150	0,483752479
12	1,112244898	0,068039879	0,075039879	5,294813878	63,53776653	150	0,42358511
15	0,711836735	0,027869135	0,034869135	3,844322082	57,66483122	150	0,384432208
18	0,494331066	0,013439976	0,020439976	3,245050612	58,41091102	150	0,389406073
20	0,400408163	0,008817968	0,015817968	3,100321796	62,00643592	150	0,413376239
22	0,330915837	0,006022791	0,013022791	3,088485121	67,94667265	150	0,452977818
25	0,256261224	0,00361184	0,01061184	3,249875949	81,24689873	150	0,541645992
28	0,204289879	0,00229539	0,00929539	3,570916835	99,98567137	150	0,666571142
30	0,177959184	0,001741821	0,008741821	3,85514302	115,6542906	150	0,771028604
35	0,130745523	0,000940192	0,007940192	4,766099974	166,8134991	150	1,112089994

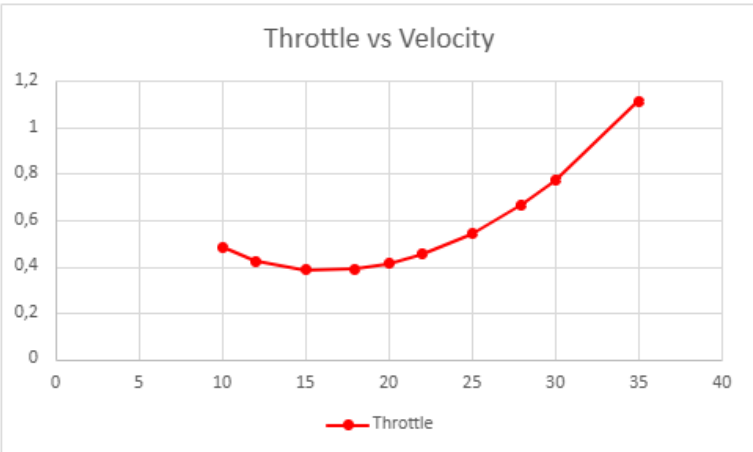

Figure 9. *Throttle vs Velocity*

As shown in Table 4 and Figure 9, the throttle value varies significantly with velocity. At low speeds, throttle starts at a relatively higher level, decreases with increasing velocity, reaches a minimum, and then increases again at higher speeds.

This behavior is explained by the variation of drag components with velocity. At low speeds, high lift requirements result in dominant induced drag, leading to higher power demand and therefore higher throttle.

As velocity increases, induced drag decreases, reducing the required power and throttle. The minimum throttle value is observed approximately at:

$$V \approx 15\, m/s$$

This speed represents the most efficient operating condition, where the engine operates at the lowest load.

At higher velocities, parasitic drag increases rapidly, causing the required power and throttle to increase. At very high speeds, throttle approaches or exceeds unity:

$$Throttle > 1$$

This indicates that the required power exceeds the available power, and steady level flight is not possible. Therefore, throttle analysis is essential for determining the maximum flight speed of the UAV.

6.3 Climb Performance Analysis

Climb performance analysis is performed to evaluate the climbing capability of the aircraft. It allows determination of how fast the aircraft can gain altitude and at which velocity the best climb performance is achieved.

Climb performance is determined using excess power, defined as:

$$P_{excess} = P_{avail} - P_{req}$$

The rate of climb (ROC) is expressed as:

$$ROC = \frac{P_{excess}}{W}$$

Table 5. *Aerodynamic coefficients, drag, power, throttle, excess power, and climb performance (ROC) values for different flight speeds*

V	CL	CDi	CD	D	P (W)	P available	Throttle	P excess	ROC
10	1,601632653	0,141087494	0,148087494	7,256287184	72,56287184	150	0,483752479	77,43712816	0,986711623
12	1,112244898	0,068039879	0,075039879	5,294813878	63,53776653	150	0,42358511	86,46223347	1,101710416
15	0,711836735	0,027869135	0,034869135	3,844322082	57,66483122	150	0,384432208	92,33516878	1,176543945
18	0,494331066	0,013439976	0,020439976	3,245050612	58,41091102	150	0,389406073	91,58908898	1,167037321
20	0,400408163	0,008817968	0,015817968	3,100321796	62,00643592	150	0,413376239	87,99356408	1,121222784
22	0,330915837	0,006022791	0,013022791	3,088485121	67,94667265	150	0,452977818	82,05332735	1,045531694
25	0,256261224	0,00361184	0,01061184	3,249875949	81,24689873	150	0,541645992	68,75310127	0,876058885
28	0,204289879	0,00229539	0,00929539	3,570916835	99,98567137	150	0,666571142	50,01432863	0,637287572
30	0,177959184	0,001741821	0,008741821	3,85514302	115,6542906	150	0,771028604	34,34570939	0,43763646
35	0,130745523	0,000940192	0,007940192	4,766099974	166,8134991	150	1,112089994	-16,8134991	-0,214239285

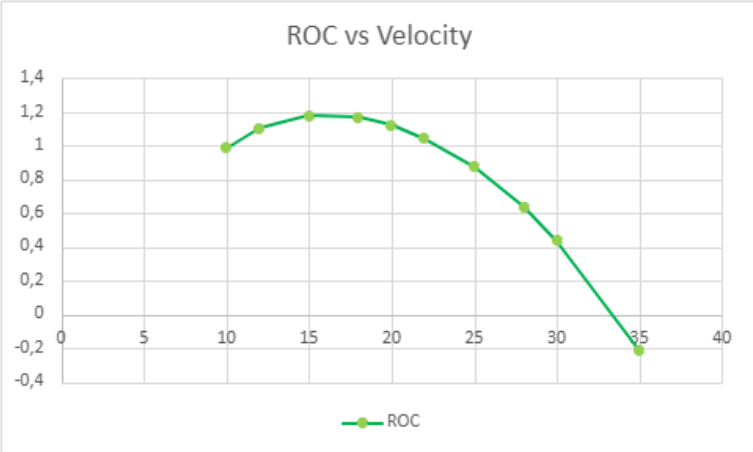

Figure 10. *Rate of Climb (ROC) vs Velocity*

As shown in Table 5 and Figure 10, the rate of climb varies significantly with velocity. At low speeds, ROC starts at a moderate value and increases with velocity, reaching a maximum in the mid-speed range.

According to the results, the maximum rate of climb is approximately:

$$ROC_{max} \approx 1.18\, m/s$$

which occurs around:

$$V \approx 15 - 18\, m/s$$

This indicates that this speed range represents the most efficient climb condition, where excess power is maximized and the engine power is most effectively converted into climb performance.

At lower speeds, high induced drag leads to increased power requirement, limiting the available excess power. Therefore, climb performance does not reach its maximum.

At higher speeds, parasitic drag increases significantly, causing power required to rise and excess power to decrease. As a result, ROC decreases.

At very high speeds, ROC approaches zero and becomes negative:

$$ROC < 0$$

This indicates that the aircraft can no longer climb and begins to lose altitude. This point corresponds to the maximum level flight speed.

7. Endurance Analysis

Endurance analysis is performed to determine how long an aircraft can remain airborne. This analysis is particularly important for missions such as surveillance, reconnaissance, and loiter operations that require long-duration flight.

For electrically powered UAVs, endurance can be expressed as:

$$E = \frac{E_{battery}}{P_{req}}$$

where E is the endurance, $E_{battery}$ is the battery energy, and P_{req} is the required power.

Table 6. *Aerodynamic, power, throttle, climb, and endurance performance parameters for different flight speeds*

V	CL	CDi	CD	D	P (W)	P avail	Throttle	P excess	ROC	Endurance(h)
10	1,601632653	0,141087494	0,148087494	7,256287184	72,56287184	150	0,483752479	77,43712816	0,986711623	4,134345739
12	1,112244898	0,068039879	0,075039879	5,294813878	63,53776653	150	0,42358511	86,46223347	1,101710416	4,721601283
15	0,711836735	0,027869135	0,034869135	3,844322082	57,66483122	150	0,384432208	92,33516878	1,176543945	5,202477726
18	0,494331066	0,013439976	0,020439976	3,245050612	58,41091102	150	0,389406073	91,58908898	1,167037321	5,136026724
20	0,400408163	0,008817968	0,015817968	3,100321796	62,00643592	150	0,413376239	87,99356408	1,121222784	4,838207447
22	0,330915837	0,006022791	0,013022791	3,088485121	67,94667265	150	0,452977818	82,05332735	1,045531694	4,415227241
25	0,256261224	0,00361184	0,01061184	3,249875949	81,24689873	150	0,541645992	68,75310127	0,876058885	3,692448631
28	0,204289879	0,00229539	0,00929539	3,570916835	99,98567137	150	0,666571142	50,01432863	0,637287572	3,00042992
30	0,177959184	0,001741821	0,008741821	3,85514302	115,6542906	150	0,771028604	34,34570939	0,43763646	2,593937487
35	0,130745523	0,000940192	0,007940192	4,766099974	166,8134991	150	1,112089994	-16,8134991	-0,214239285	1,798415606

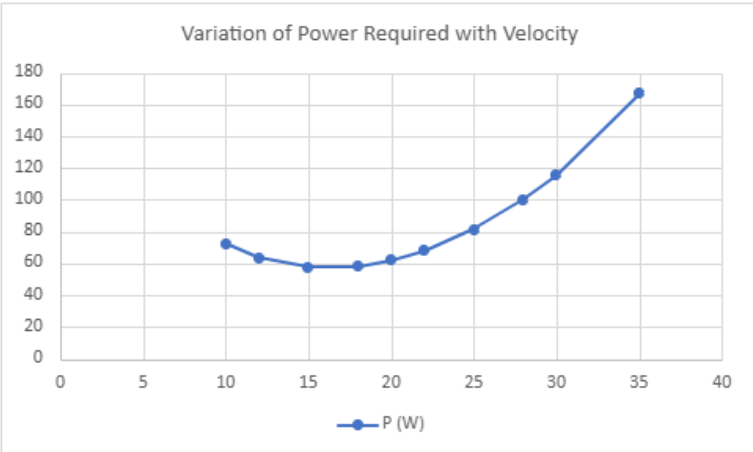

Figure 11. *Variation of Power Required with Velocity*

As shown in Figure 11, the required power initially decreases with increasing velocity and then increases after a certain point. This behavior is explained by the dominance of induced drag at low speeds and parasitic drag at higher speeds.

According to the results, the minimum required power is approximately:

$$P_{min} \approx 57.6\, W$$

which occurs at approximately:

$$V \approx 15\, m/s$$

This condition represents the maximum endurance point of the aircraft. At this speed, energy consumption is minimized, allowing the UAV to remain airborne for the longest possible duration.

At lower speeds, induced drag increases due to higher lift requirements, leading to higher power consumption and reduced endurance. Similarly, at higher speeds, the increase in parasitic drag results in higher power demand, reducing the endurance.

Assuming a battery capacity of **300 Wh**, the maximum endurance can be estimated as:

$$E_{max} = \frac{300}{57.6} \approx 5.21\, hours$$

This result indicates that the UAV can achieve a relatively long flight duration at the optimal speed.

The maximum endurance point is particularly important for missions such as surveillance, reconnaissance, and loitering, where long flight duration is required.

However, this value represents a theoretical maximum. In real flight conditions, factors such as battery efficiency, system losses, and safety margins will reduce the achievable endurance.

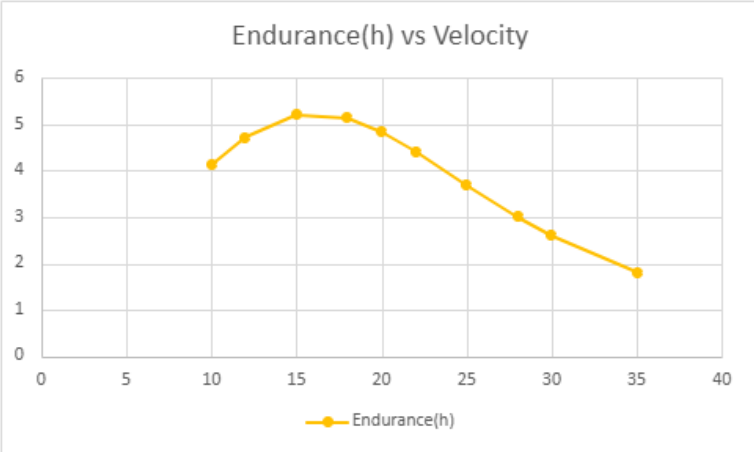

Figure 12. *Endurance vs Velocity*

As shown in Figure 12, the endurance initially increases with velocity, reaches a maximum at a certain point, and then decreases as velocity continues to increase.

This behavior is directly related to the variation of power required with velocity. At low speeds, high induced drag leads to increased power consumption, resulting in lower endurance.

As velocity increases, induced drag decreases and power required reaches a minimum. At this point, energy consumption is minimized, and endurance reaches its maximum value.

At higher speeds, parasitic drag increases significantly, causing power consumption to rise and endurance to decrease.

8. Range Analysis

Range analysis is performed to determine the maximum distance an aircraft can travel with a given energy capacity. This analysis is particularly important for long-range missions and flight planning.

For electrically powered UAVs, range is expressed as:

$$Range = V \times Endurance$$

In this study, range values were calculated using the endurance values obtained for different velocities, and the results are presented in Table 6 and Figure 13.

Table 7. *Aerodynamic, power, throttle, climb, endurance, and range performance parameters for different flight speeds*

V	CL	CDi	CD	D	P (W)	P avail	Throttle	P excess	ROC	Endurance(h)	Range(km)
10	1,601632653	0,141087494	0,148087494	7,256287184	72,56287184	150	0,483752479	77,43712816	0,986711623	4,134345739	148,8364466
12	1,112244898	0,068039879	0,075039879	5,294813878	63,53776653	150	0,42358511	86,46223347	1,101710416	4,721601283	203,9731754
15	0,711836735	0,027869135	0,034869135	3,844322082	57,66483122	150	0,384432208	92,33516878	1,176543945	5,202477726	280,9337972
18	0,494331066	0,013439976	0,020439976	3,245050612	58,41091102	150	0,389406073	91,58908898	1,167037321	5,136026724	332,8145317
20	0,400408163	0,008817968	0,015817968	3,100321796	62,00643592	150	0,413376239	87,99356408	1,121222784	4,838207447	348,3509362
22	0,330915837	0,006022791	0,013022791	3,088485121	67,94667265	150	0,452977818	82,05332735	1,045531694	4,415227241	349,6859974
25	0,256261224	0,00361184	0,01061184	3,249875949	81,24689873	150	0,541645992	68,75310127	0,876058885	3,692448631	332,3203768
28	0,204289879	0,00229539	0,00929539	3,570916835	99,98567137	150	0,666571142	50,01432863	0,637287572	3,00042992	302,443336
30	0,177959184	0,001741821	0,008741821	3,85514302	115,6542906	150	0,771028604	34,34570939	0,43763646	2,593937487	280,1452486
35	0,130745523	0,000940192	0,007940192	4,766099974	166,8134991	150	1,112089994	-16,8134991	-0,214239285	1,798415606	226,6003663

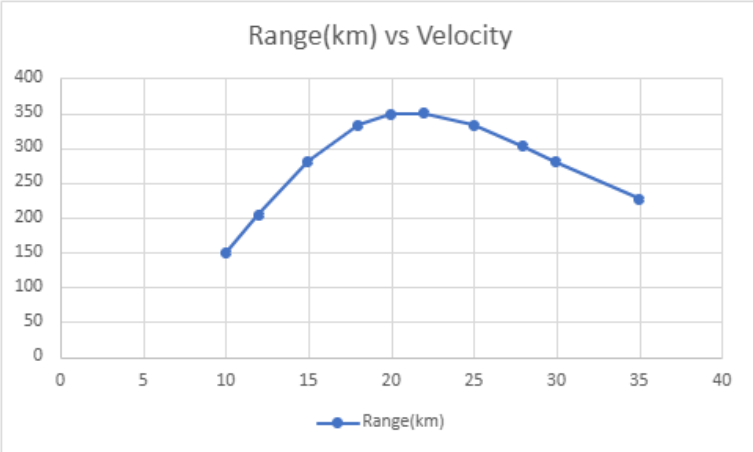

Figure 13. *Range(km) vs Velocity*

As shown in Table 7 and Figure 13, the range initially increases with velocity, reaches a maximum at a certain point, and then decreases as velocity increases further. This behavior is explained by the variation of aerodynamic drag components with velocity.

According to the results, the maximum range is approximately:

$$Range_{max} \approx 349.7\ km$$

which occurs at approximately:

$$V \approx 22\ m/s$$

This speed corresponds to the condition of minimum drag and maximum aerodynamic efficiency. At this point, the aircraft travels the maximum distance per unit of energy.

At low speeds, high induced drag increases energy consumption and limits the range. At high speeds, increased parasitic drag raises power consumption and reduces endurance, leading to a decrease in range.

The results indicate that the optimal velocity for maximum range is approximately **22 m/s**. Operating at this speed allows the UAV to achieve maximum distance with the available energy.

This confirms the strong relationship between the minimum drag condition and maximum range. Therefore, selecting this velocity during mission planning significantly improves operational efficiency.

9. Flight Envelope

The flight envelope defines the velocity range in which the aircraft can operate safely and sustain flight. This envelope is determined by evaluating the main performance limits such as stall speed, minimum safe flight speed, maximum endurance speed, maximum range speed, and maximum level flight speed.

In this study, the flight envelope of the UAV was determined by combining the results of stall, drag, power required, power available, endurance, and range analyses. According to the results, the stall speed was found to be approximately:

$$V_{stall} \approx 11\, m/s$$

The minimum safe operational speed was taken as approximately 20% above the stall speed:

$$V_{safe} \approx 13\, m/s$$

Based on the power required and power available analyses, the maximum level flight speed was determined as approximately:

$$V_{max} \approx 33\, m/s$$

Beyond this point, the required power exceeds the available power, and steady level flight is no longer possible.

In addition, the maximum endurance speed was found to be approximately:

$$V_{endurance} \approx 15\, m/s$$

while the maximum range speed was obtained as:

$$V_{range} \approx 22\, m/s$$

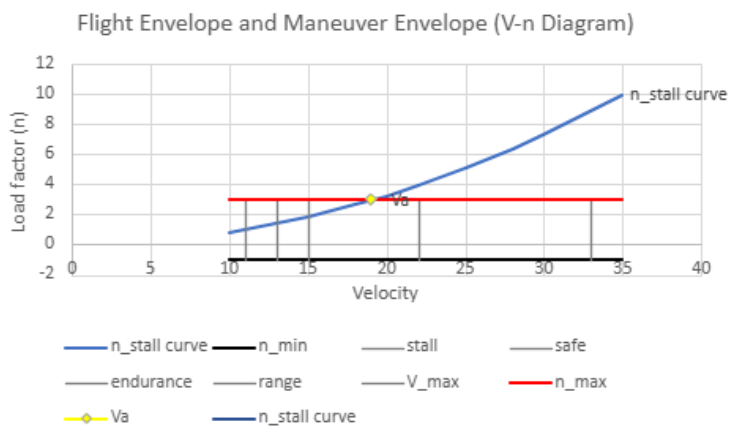

Figure 14. *Flight Envelope and Maneuver Envelope (V-n Diagram)*

Figure 14 shows the stall boundary, the structural load limit, and the maneuver speed together. The stall boundary defines the low-speed limit of the aircraft due to insufficient lift, while the horizontal structural limit line indicates the maximum allowable load factor. The intersection of these two boundaries is defined as the maneuver speed and is calculated using the following equation.

$$V_a = \sqrt{\frac{2Wn}{\rho S C_{L_{max}}}}$$

In this study:

- $W = 78.48\ N$
- $n_{max} = 3$
- $\rho = 1.225\ kg/m^3$
- $S = 0.8\ m^2$
- $C_{L_{max}} = 1.3$

The maneuver speed has been calculated using these values.

$$V_a \approx 19.2\ m/s$$

This speed represents the critical point at which the stall boundary intersects with the structural limit.

According to these results, the safe flight envelope of the UAV can be defined approximately as:

$$13\ m/s \leq V \leq 33\ m/s$$

Within this velocity range, the aircraft can sustain safe flight; it achieves maximum endurance at approximately 15 m/s and maximum range at approximately 22 m/s. Therefore, the flight envelope not only defines the velocity range in which flight is possible, but also identifies the most suitable operating regions depending on mission requirements.

10. Conclusion

In this study, the flight performance of an unmanned aerial vehicle (UAV) has been comprehensively analyzed. The analysis included aerodynamic characteristics, drag force, power required and available, throttle behavior, climb performance, endurance, and range.

The results indicate that the stall speed of the aircraft is approximately 11 m/s, and the minimum safe flight speed is about 13 m/s. Based on the power required and power available analysis, the maximum flight speed is approximately 33 m/s.

The endurance analysis shows that the UAV achieves maximum flight duration at approximately 15 m/s. The range analysis indicates that the maximum range is approximately 349 km at around 22 m/s. These results are consistent with the minimum power and minimum drag conditions and align well with theoretical expectations.

Within the maneuver envelope (V-n diagram), the maximum load factor is assumed to be 3, and the maneuver speed is calculated as approximately 19.2 m/s. This speed represents the critical point at which the aircraft can achieve maximum maneuverability without exceeding structural limits.

The flight envelope analysis defines the safe operational velocity range of the UAV as approximately 13 m/s to 33 m/s. Within this range, optimal flight conditions for different mission profiles have been identified.

In conclusion, the analyses conducted in this study successfully define the aerodynamic, performance, and structural limits of the UAV and provide a comprehensive understanding of its safe and efficient operating conditions. This work can serve as a valuable reference for the design and performance evaluation of similar aerial platforms.